Consciousness, Sleep and Attention

What attention can and can't do, how sleep works, what losing it costs, and what dreams, hypnosis and everyday drugs do to the mind.

Isagi · November 2026 · Spine: Ciccarelli & White, *Psychology* (4th ed.), Ch. 4 — rewritten, updated and applied

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Field sales runs on two scarce resources nobody budgets for: attention and sleep. A territory manager drives 300 km, answers forty WhatsApp messages, sits in three dealer meetings and gets home at ten. Every decision in that day draws on an attention system that can only do one demanding thing at a time, and on a brain that degrades predictably when it's short of sleep. This part explains both — and why the "I'm fine on five hours" manager is usually wrong about himself.

1 What consciousness is

Consciousness is your awareness of yourself and your surroundings at a given moment: what you're thinking, feeling and perceiving. **Waking consciousness** is the ordinary clear, organised state. **Altered states** are shifts in the quality or pattern of awareness — sleep, dreaming, daydreaming, hypnosis, meditation, and the effects of drugs. Much of the brain's work never reaches consciousness at all: you don't experience the calculations that keep you balanced or that turn a shape into a word. Consciousness is more like a spotlight than a room.

2 Attention: a narrow spotlight

Selective attention is the ability to focus on one input and filter out the rest. The classic example is the **cocktail-party effect**: in a noisy room you follow one conversation and tune out the others — yet you'll still notice if someone across the room says your name. The filter isn't a wall; it lets through things that matter to you.

The cost of a narrow spotlight is that we miss things we assume we'd see. In a famous 1999 study by Daniel Simons and Christopher Chabris, people counted basketball passes in a video. About half failed to notice a person in a gorilla suit who walked through the middle of the game and thumped their chest. This is **inattentional blindness**: failing to see something in plain view because attention is elsewhere. Its cousin, **change blindness**, is failing to notice a change between one view and the next.

The multitasking myth

For anything that needs thought, the brain doesn't run tasks in parallel; it switches between them. Each switch has a cost in time and errors, and the costs are largest for complex or unfamiliar tasks. People who believe they multitask well tend to perform no better, and in some studies worse. Two tasks combine smoothly only if one is so practised that it's nearly automatic — walking while talking, not drafting an email during a negotiation.

The most serious case is the phone while driving. A hands-free kit frees the hands but not the attention; the conversation still competes for the same limited capacity. For a sales force that spends half its working life on the road, "no calls while driving" is a safety rule, not a courtesy.

3 The body clock

Circadian rhythms are biological cycles of roughly 24 hours — body temperature, alertness, hormone levels. The master clock is a small cluster of cells in the hypothalamus, the **suprachiasmatic nucleus**,

which is reset daily by light reaching the eyes. As evening light fades, the clock signals the pineal gland to release **melatonin**, which makes you sleepy. Bright screens late at night delay that signal.

Two features matter at work. Most people have a natural dip in alertness in the early afternoon — the post-lunch slump is mostly the clock, not the lunch. And the clock adjusts slowly: jet lag and rotating night shifts leave people working while their body expects sleep, with more errors and accidents in those hours.

4 The stages of sleep

Using the EEG (Part 1), researchers found that sleep is not one state but a repeating cycle:

- **N1** — light sleep, a few minutes; easily woken; sometimes a sudden jerk.
- **N2** — true sleep; bursts of activity called sleep spindles; about half the night is spent here.
- **N3** — deep, **slow-wave sleep**; hardest to wake from; the body's physical repair and growth-hormone release are concentrated here.
- **REM** (rapid eye movement) sleep — the brain is highly active, the eyes dart, and most vivid dreams occur. The body's major muscles are temporarily paralysed, so you don't act out your dreams.

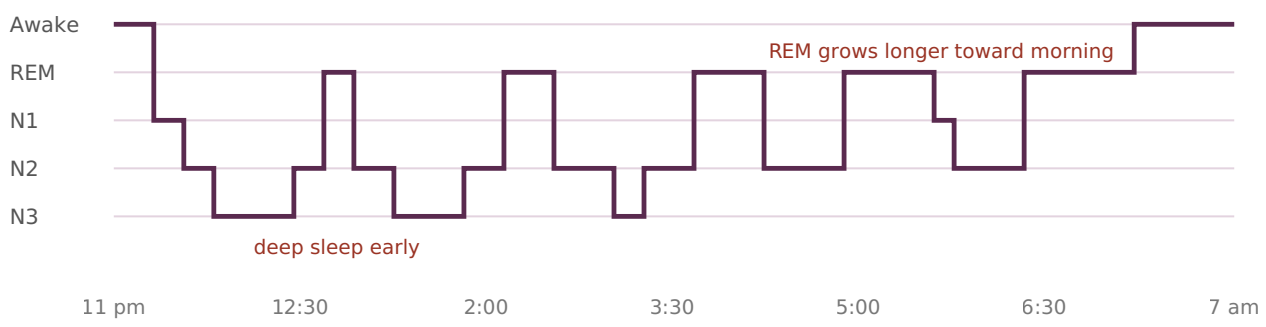


Figure 1. A simplified night (a "hypnogram"). Cycles last roughly 90 minutes. Deep N3 sleep is concentrated early in the night and REM late, which is why cutting the last two hours of sleep removes a disproportionate share of REM.

5 What sleep loss costs

Why we sleep has several answers, all partly right: to conserve energy and stay safe at night (the **adaptive** view), to repair the body and clear the brain's waste products (the **restorative** view), and to consolidate the day's learning into long-term memory. Whatever the reason, the cost of skipping it is well measured.

In a 2003 laboratory study by Hans Van Dongen and colleagues, people restricted to six hours a night for two weeks declined steadily on attention tests until they were as impaired as people kept awake for one to two nights straight. The most important finding: the six-hour group rated themselves as only slightly sleepy. Chronic sleep loss impairs the judgement you would use to notice it.

Other studies have compared fatigue with alcohol. After about 17–19 hours awake — up at 6 am, still driving home at midnight — performance on some tests falls to the level seen at a blood-alcohol concentration of about 0.05%. India's legal driving limit is 0.03%.

WORKED EXAMPLE 1 · THE MICROSLEEP

A tired driver can fall into a **microsleep**: a lapse of a few seconds, often with eyes open, that the driver may not even notice. How far does the car travel during a 3-second microsleep?

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60 km/h = 60,000 m / 3,600 s ≈ 16.7 m/s
3 s × 16.7 m/s ≈ 50 m

80 km/h ≈ 22.2 m/s → 3 s × 22.2 ≈ 67 m
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Fifty to seventy metres — two and a half to three cricket pitches — with nobody steering. That is the case for a company rule capping driving after a long day, or for paying for a night's stay over pushing a rep to drive home.

WORKED EXAMPLE 2 · SLEEP DEBT OVER A WEEK

A manager needs about 8 hours but sleeps 6 on weeknights, then 9 on Saturday and Sunday.

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Weekday shortfall: 5 nights × (8 − 6) h = 10 h
Weekend payback:   2 nights × (9 − 8) h =  2 h
Net debt carried into Monday            =  8 h
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A weekend lie-in recovers some alertness, but studies of "catch-up" sleep show it does not fully reverse the effects of a week of restriction, and the pattern itself disturbs the body clock. Regular hours beat recovery binges.

6 Sleep disorders

- **Insomnia** — persistent trouble falling or staying asleep. The most effective long-term treatment is a structured talking therapy, cognitive behavioural therapy for insomnia (CBT-I), not sleeping pills.
- **Obstructive sleep apnoea** — the airway repeatedly closes during sleep, so breathing stops briefly many times an hour. Signs include loud snoring, gasping and heavy daytime sleepiness. It is common, underdiagnosed and a known risk factor for road accidents among professional drivers. It is treatable.
- **Narcolepsy** — sudden, uncontrollable attacks of sleep during the day, sometimes going straight into REM.
- **Sleepwalking** and **night terrors** happen in deep N3 sleep and are usually not remembered; **nightmares** are bad dreams in REM and usually are.

7 Dreams

Freud saw dreams as disguised wish-fulfilment, with a hidden meaning to decode. There is little scientific support for that. The **activation-synthesis** theory (Hobson and McCarley, 1977) proposed that during REM the brainstem sends random signals and the cortex stitches them

into a story. Later views add that dreams draw heavily on the day's concerns and may help process emotions and memories. The honest summary: we know when dreams happen better than why.

8 Hypnosis

Hypnosis is a state of focused attention in which a person becomes more responsive to suggestion. It is real in the sense that people's experience changes — it can reduce how much pain bothers someone, which is why it has a role alongside medical treatment. People differ widely in how hypnotisable they are. Two views compete: that hypnosis splits awareness into separate streams (**dissociation**), or that it is people playing the role of a hypnotised person in response to expectations (**social-cognitive** theory). What it cannot do is reliably recover hidden memories; it makes people more confident in what they recall, including things that never happened (Part 5).

9 Psychoactive drugs

Psychoactive drugs alter thinking, perception, mood or behaviour. Repeated use can bring **tolerance** (needing more for the same effect), **withdrawal** (unpleasant symptoms on stopping), and **dependence**, which can be physical, psychological or both. The main groups:

- **Stimulants** speed up the nervous system: caffeine, nicotine, and illegal drugs such as cocaine and amphetamines.
- **Depressants** slow it down: alcohol, and sedatives. Alcohol first reduces inhibition, which is why it feels like a stimulant at a party, and in larger amounts impairs judgement, coordination and memory.
- **Opioids** relieve pain and produce euphoria; they carry a high risk of dependence.
- **Hallucinogens** alter perception, for example LSD; cannabis has mixed effects.

The workplace version is mostly legal drugs. Caffeine masks sleepiness without repaying sleep debt, and late in the day it delays sleep further. Alcohol is built into many sales cultures through dealer meets and

channel parties; it helps people fall asleep but fragments the second half of the night, cutting REM. Hosting the party is a business choice. Making drinking part of how people are judged is a culture problem.

10 Held up?

Claim	Verdict	Why
"I can multitask"	**No**	The brain switches, with costs. Only highly automatic tasks combine well.
"I'm fine on five or six hours"	**Rarely**	True short sleepers exist but are rare. Most restricted sleepers are impaired and don't feel it.
"Weekend lie-ins cancel weekday debt"	**Partly**	Some recovery, not full.
"You can learn new material while asleep"	**No**	Sleep strengthens what you learned while awake; you can't absorb new complex information while asleep.
"Hypnosis recovers buried memories"	**No**	It raises confidence, not accuracy.
A short nap restores alertness	**Yes**	Naps of about 10–20 minutes improve alertness without deep-sleep grogginess.

Summary

- **Consciousness** is moment-to-moment awareness; sleep, dreams, hypnosis and drugs are **altered states**. Most brain processing never becomes conscious.
- **Attention** is a narrow spotlight. **Inattentional blindness** means we miss things in plain view. Multitasking is really task-switching, with costs; phone calls while driving are dangerous even hands-free.
- The **circadian clock** (suprachiasmatic nucleus, melatonin, light) runs on about 24 hours and explains the afternoon dip, jet lag and shift-work errors.
- Sleep cycles roughly every 90 minutes through **N1, N2, N3** (deep) and **REM** (dreams, muscle paralysis). Deep sleep comes early; REM late.
- Six hours a night for two weeks impairs attention as much as one to two nights without sleep, and people don't notice. Long hours awake impair like alcohol. A 3-second microsleep at 60 km/h is 50 m unsteered.
- **Sleep apnoea** is common and a road-safety risk; **insomnia** is best treated with CBT-I.
- Dream theories range from Freud (little support) to **activation-synthesis**.
- **Hypnosis** raises suggestibility and can ease pain, but doesn't recover memories.
- Drugs are stimulants, depressants, opioids or hallucinogens; tolerance, withdrawal and dependence follow repeated use. Caffeine masks sleep debt; alcohol disrupts sleep.

Practice problems

P1. A rep has a 2-second microsleep at 90 km/h. How far does the car travel?

P2. A manager needs 7.5 hours but sleeps 5.5 on five weeknights and 8.5 on two weekend nights. What is the net sleep debt at the end of the week?

P3. During a quarterly review, the presenting manager misses that the chart on screen has the wrong month, though everyone else saw it. Which concept explains this? What process change would help?

P4. A senior colleague says he "only needs five hours" and is sharper than anyone. What does the 2003 restriction study suggest about his self-assessment?

P5. A driver on your team snores loudly, gasps at night and nods off at red lights. What condition might this suggest, and what should you do as his manager?

P6. Write a three-point travel and phone policy for a field sales team, using this part's evidence.

Glossary

Consciousness

Awareness of oneself and one's surroundings at a given moment.

Altered state

A shift in the quality or pattern of consciousness.

Selective attention

Focusing on one input while filtering out others.

Inattentional blindness

Failing to notice something in plain view because attention is elsewhere.

Task-switching cost

Time and errors lost each time attention shifts between tasks.

Circadian rhythm

A biological cycle of about 24 hours.

Melatonin

A hormone released in darkness that promotes sleep.

N3 (slow-wave) sleep

The deepest stage of non-REM sleep.

REM sleep

Sleep with rapid eye movements, vivid dreams and muscle paralysis.

Microsleep

A brief, involuntary lapse into sleep lasting a few seconds.

Sleep apnoea

Repeated pauses in breathing during sleep.

Activation-synthesis

The theory that dreams are the cortex making sense of random brainstem signals.

Hypnosis

Focused attention with heightened responsiveness to suggestion.

Tolerance / withdrawal

Needing more of a drug for the same effect / symptoms on stopping.

Stimulant / depressant
A drug that speeds up / slows down the nervous system.

Answer key

P1. 90 km/h = 25 m/s. 2 s × 25 = 50 m.

P2. Shortfall 5 × 2 = 10 h. Payback 2 × 1 = 2 h. Net debt = 8 h.

P3. Inattentional blindness: his attention is on delivery, not on checking the slide. Fix: someone other than the presenter checks the deck before the meeting.

P4. People restricted to six hours rated themselves only slightly sleepy while their measured performance fell sharply. Self-assessment is the least reliable evidence; measured performance is what counts.

P5. These are classic signs of obstructive sleep apnoea. Don't diagnose; suggest he see a doctor, and keep him off long drives until he has. It's treatable.

P6. For example: (1) no calls while driving, including hands-free — pull over or call back; (2) no driving home after a working day beyond a set limit — book a stay; (3) no expectation of replies to late-evening messages.

Sources: chapter structure follows Ciccarelli & White, *Psychology*, 4th ed. (Pearson), Ch. 4; text and figures are original. Inattentional blindness: Simons & Chabris, *Perception*, 1999. Sleep restriction: Van Dongen et al., *Sleep*, 2003. Fatigue and alcohol: Williamson & Feyer, *Occupational and Environmental Medicine*, 2000. Activation-synthesis: Hobson & McCarley, *American Journal of Psychiatry*, 1977. India's limit of 30 mg per 100 ml blood: Motor Vehicles Act, s. 185.